F172 Lesson 12: CRISPR-Cas9 Gene Editing & Research Workshop

OCR AAQ Level 3 Human Biology · Unit F172: Genetics (Dan)

Molecular Engine | PAM & sgRNA | NHEJ vs HDR | NEA Distinction Workshop

Workshop Goal: Master the molecular mechanisms of CRISPR-Cas9 and develop advanced evaluative research skills for Genetics NEA Task 2.

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Biological Origins: Adaptive Bacterial Immunity



1. CRISPR Loci

Clustered Regularly Interspaced Short Palindromic Repeats found in bacterial and archaeal genomes.



2. Viral Memory

Bacteria capture foreign bacteriophage DNA fragments and store them as spacers between repeats.

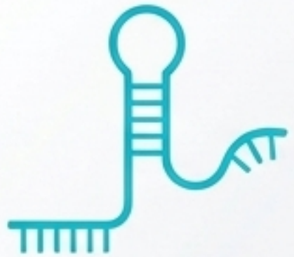


3. Adaptive Defense

Transcribed crRNA guides Cas endonucleases to recognize and cleave invading bacteriophage genomes upon re-infection.

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The CRISPR-Cas9 Engine: 3 Essential Components



Single Guide RNA (sgRNA)

Synthetic 20-nucleotide spacer engineered to match target genomic sequence, fused to structural tracrRNA scaffold.



Protospacer Adjacent Motif (PAM)

Essential 5′-NGG-3′ DNA sequence immediately adjacent to target site; physical prerequisite for Cas9 binding.



Cas9 Endonuclease

Multi-domain enzyme with RuvC and HNH catalytic nuclease domains that cleave opposing strands, creating a targeted double-strand break (DSB).

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Step-by-Step Mechanism of Cas9 Target Cleavage

1. PAM Interrogation:
Cas9 scans the genome, binding PAM (5'-NGG-3') motifs through protein-DNA interactions.



2. DNA Unwinding:
Cas9 unwinds adjacent target DNA, allowing sgRNA to test for complementary base pairing.

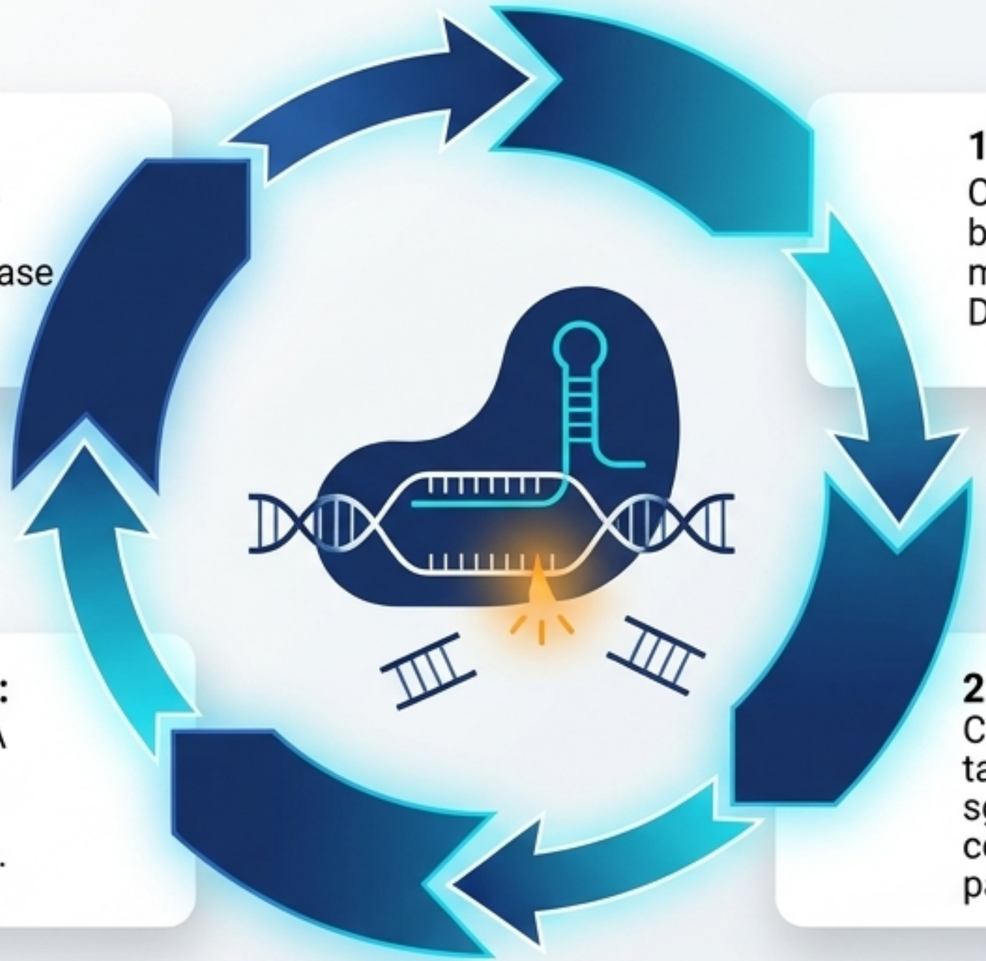


3. R-Loop Formation:
Stable 20 bp RNA:DNA hybrid forms between sgRNA spacer and genomic target strand.



4. Targeted Cleavage:
RuvC and HNH domains induce a blunt double-strand break exactly 3 base pairs upstream of PAM.





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DNA Repair Pathway 1: NHEJ Gene Knockout

Mechanism:
Error-prone direct re-ligation of broken blunt DNA ends without an homologous repair template.

Mutation Outcome:
Frequent introduction of random insertions or deletions (indels) at the cleavage site.

Therapeutic Application:
Induces reading frameshift mutations and premature stop codons, knocking out pathogenic dominant genes (e.g. viral receptors, oncogenes, or mHTT).



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DNA Repair Pathway 2: HDR Gene Correction



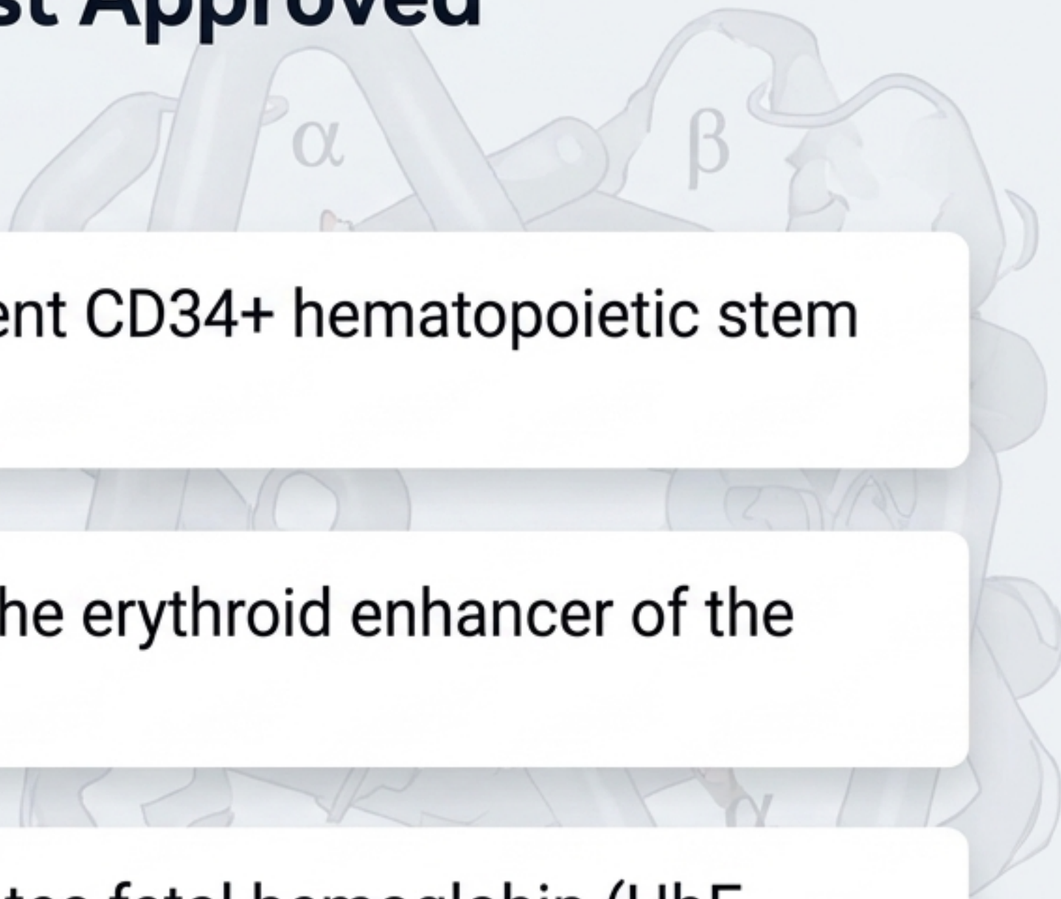
Mechanism: High-fidelity template-dependent repair pathway that utilizes an exogenous synthetic donor DNA template.

Precision Outcome: Enables exact nucleotide substitution, point mutation correction, or therapeutic sequence knock-in without indels.

Clinical Limitations: Primarily active in late S and G2 phases of dividing cells; inefficient in non-dividing post-mitotic cells (e.g. adult neurons).

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Clinical Landmark: Casgevy (First Approved CRISPR Therapy)



Therapeutic Target: Ex vivo editing of patient CD34+ hematopoietic stem and progenitor cells (HSPCs).

Mechanism: CRISPR-Cas9 NHEJ disrupts the erythroid enhancer of the BCL11A gene.

Clinical Benefit: Silencing BCL11A reactivates fetal hemoglobin (HbF, alpha2-gamma2), preventing sickle hemoglobin polymerization and eliminating vaso-occlusive pain crises.

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Technical Obstacles: Off-Target Cleavage & Delivery

Off-Target Effects

Cas9 can tolerate 1–3 base mismatches between sgRNA and non-target genomic sites, risking oncogene activation or chromosomal translocations.

p53 DNA Damage Response

Double-strand breaks activate p53-mediated apoptosis, potentially selecting for cells with preexisting p53 mutations (increasing cancer risk).

Delivery Bottlenecks

Packaging large Cas9/sgRNA complexes into effective viral vectors (AAV) remains difficult in vivo due to payload size limits and immune clearance.

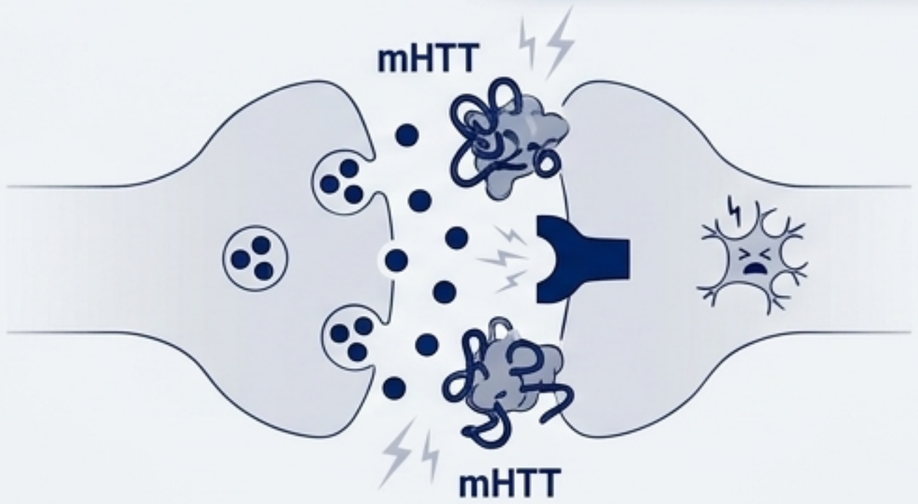
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NEA Task 2 Case Study: Targeting Huntington's Disease (HD)

The Genetic Defect

Pathogenic Defect:
Autosomal dominant expansion of CAG repeats in the HTT gene (IT15).

Cellular Outcome:
Produces mutant huntingtin (mHTT) clumps leading to discontinued neurotransmission and cellular toxicity.



The Therapeutic Strategy

Delivery Strategy:
Utilizing Adeno-Associated Virus (AAV) vectors targeting the central nervous system.

CRISPR Objective:
Induce NHEJ to selectively silence the mutant HTT allele while preserving normal cellular function.



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NEA Task 2: Distinction-Level Evaluation Framework

Medical Benefits [M3] [P5]

- Curative potential for monogenic diseases.
- Targeted molecular precision over broad-spectrum symptom management.

Technical Limitations [D2]

- Risk of off-target mutations.
- Immune responses to AAV delivery vectors.
- Extreme inefficiency in post-mitotic adult neurons.

Ethical Considerations [D2]

- Somatic vs. Germline editing debates.
- Extreme high costs (e.g., Casgevy) limiting equity of access.
- Unforeseen long-term genetic consequences.

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